

Education

East China Normal University

Sep. 2023 – Jun. 2027

B.S. in Computer Science, Elite Computer Science Program

- GPA: **3.84/4.00**; ranked **2/25**.
- Awards: **National Scholarship**; ECNU Outstanding Student Scholarship and Outstanding Student Award.
- English: TOEFL **106**; CET-6 **616**; CET-4 **657**.
- Selected coursework: Large Language Models (Honors), AI, NLP, Computer Vision, Probability and Statistics, Operating Systems, Computer Networks, Databases, and Compilers; highest grade in all listed courses.

Research Experience

Rewriting the Vision-Language Bridge: Fast Closed-Form Backdoor Editing in MLLM Connectors

Apr. 2026 – Present

*Trustworthy AI Security / First-Author Manuscript in Preparation for AAAI 2027***Keywords:** MLLM Backdoor, Closed-Form Model Editing, Stealthy Visual Triggers, Leakage Suppression

- Lead the development of a training-free closed-form connector-editing method that casts backdoor injection as a constrained mapping from visual features to target semantic representations in MLLMs.
- Designed corner-fragment and edge-cropped visual triggers, then built a projected-feature screening pipeline to balance targeted induction strength, visual stealthiness, and false-trigger risk.
- Introduced Fisher-discriminant-based column-wise gating to softly weight edited connector dimensions and suppress semantic leakage, reducing the false-trigger rate from **37% to 18%**.
- Achieved approximately **92% targeted attack success** while retaining **85% clean accuracy** on LLaVA-v1.5-7B using only a small background set.

LLM Agents, Multimodal Reliability, and Collaborative Reasoning

Jun. 2026 – Present

*Research Intern, Shanghai Artificial Intelligence Laboratory***Keywords:** Data Agent, VLM Hallucination, Confidence Calibration, Collaborative Reasoning

- Investigate natural-language-to-analysis workflows for the **DataElf** data agent, spanning task understanding, data querying, tool use, result verification, and report generation.
- Study hallucination and confidence calibration in multimodal models, including uncertainty estimation, answer verification, and failure analysis for visual-question answering and multimodal reasoning.
- Explore knowledge–reasoning decoupling for collaborative LLM reasoning, with emphasis on self-consistency, step-level knowledge injection, and evidence-aware intermediate states.

Relation-Aware Multi-Agent Collaboration with Long-Term Memory

Apr. 2026 – Jun. 2026

*Research Intern, Central Research Institute, Fudan Microelectronics Group***Keywords:** Multi-Agent Collaboration, A2A Communication, Tool Governance, Long-Term Memory

- Designed a directed-graph A2A architecture with structured handoff envelopes, asynchronous evidence exchange, provenance-aware synthesis, and relationship-sensitive task delegation.
- Added tool-access governance to prevent private memory and workspace content from being published through unauthorized channels.
- Built a DDR-Benchmark 10-K evaluation pipeline with message-level and trajectory-level metrics; the system achieved **76.48% message-wise** and **53.66% trajectory-wise accuracy**.

DialecticFlow: Long-Horizon Strategy Control for Chinese Debate Agents

Dec. 2025 – Jan. 2026

*LLM Fine-Tuning and Agent Systems / Project Lead / Ranked 1st***Keywords:** Qwen2.5-7B, LoRA, Stance-Aware RAG, Dual-Track Memory

- Built a controllable Qwen2.5-7B debate agent with **dual-track memory**, stance-aware RAG, post-generation verification, and diagnostic rewriting.
- Curated **29,244** instruction samples and sequentially trained LoRA adapters for structure, real-debater style, and adversarial robustness.
- Conducted five-way ablations; the full system improved the normalized average score from **29.5 to 94.0** across logic, rebuttal, stance consistency, closing quality, and compliance.

Two-Stage Diffusion Purification for Backdoor Defense

Dec. 2024 – Mar. 2025

*Trustworthy AI / Student Inventor, Chinese Invention Patent No. 202511807335.9***Keywords:** Diffusion Purification, Backdoor Defense, SSIM Localization, Patent

- Developed a black-box defense combining global DDPM purification, SSIM-guided trigger localization, and adaptive local denoising.
- Reduced attack success from over 90% to below **6.15%** on CIFAR-10 and GTSRB while retaining at least **90% clean accuracy**; patent under substantive examination.

Selected Projects

Six-Axis Painting Robot on SylixOS and EtherCAT

Apr. 2025 – Jun. 2026

Embodied AI and Real-Time Systems / Core Member / National Innovation Project

Keywords: SylixOS, EtherCAT, Diffusion Models, Robotic Arm

- Ported the EtherCAT stack and adapted network drivers on SylixOS; contributed to image-to-stroke conversion, trajectory planning, and robotic-arm execution.
- Helped integrate the real-time control chain from generated strokes to mechanical-arm motion. The project won the **National Third Prize** in the 2025 National Student Computer System Capability Competition.

Building Large Language Model Systems

Sep. 2025 – Dec. 2025

Independent Implementation Based on Stanford CS336 / A1 Rank 14/70

Keywords: LLM Pretraining, FlashAttention, Parallelism, Data Filtering

- Implemented BPE tokenization, Transformer training, FlashAttention-2-style forward/backward passes, tensor and pipeline parallelism, and MinHash-LSH data filtering.
- Reproduced key components of modern LLM training systems from scratch and ranked **14/70** on the A1 global leaderboard.

MeteoNet: Multi-Model Weather Forecasting and Visualization

Apr. 2025 – Jun. 2025

Individual Foundations of AI Course Project

Keywords: ERA5, Spatiotemporal Forecasting, Spatial Attention, Mixed-Precision Training

- Compared U-Net, Pangu-Weather, FengWu, and a custom model on ERA5; designed residual spatial-attention blocks, depthwise-separable convolution, multi-step prediction, and weighted objectives.
- Built visualization and evaluation modules for multi-variable weather forecasting, supporting spatial error analysis and model comparison across prediction horizons.

MySmartFrig: Multimodal Food Management and Recipe Generation

2025 – 2026

Independent NLP Application / Published on the Apple App Store

Keywords: Multimodal Extraction, Recipe Generation, Ionic Vue, Django

- Independently designed, developed, tested, and released a full-stack iOS application for image- and speech-based ingredient extraction, inventory management, and contextual recipe generation.
- Implemented the mobile frontend, backend APIs, database schema, and App Store release workflow using Ionic Vue, Capacitor, Django, and MySQL.

Smart Classroom: AI Assessment for Teacher Training

2024 – 2025

National Undergraduate Innovation Project / Team Member

Keywords: AI Teaching Analytics, Digital Human, Classroom Skill Assessment

- Contributed to an intelligent teaching-analysis platform and an AI digital-human instruction system for evaluating and improving pre-service teachers' classroom skills.

Efficient Model Training for Cyber-Physical Systems

2025 – 2026

Shanghai Municipal Undergraduate Innovation Project / Team Member

Keywords: Federated Learning, Non-IID Data, Privacy Protection, Edge Devices

- Studied federated learning under non-IID data, privacy constraints, and heterogeneous edge devices; contributed to an efficient collaborative-training framework for cyber-physical environments.

Teaching, Service & Activities

Teaching Assistant, Cloud Computing Systems

Jul. 2026 – Present

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- Facilitate laboratories, provide technical guidance, grade assignments and reports, answer questions, and support course administration.

Teaching Assistant, Computer Networks

Sep. 2025 – Jan. 2026

East China Normal University

- Led networking laboratories, graded assignments and reports, answered technical questions, and assessed presentations and oral examinations.

Service and Activities

2023 – Present

East China Normal University

- Served as student mentor in the School of Data Science and Engineering and member of the External Relations Department; volunteered at the 113th OSC Shanghai event and delivered a Young Marxists Training Programme talk.
- Joined an academic exchange programme at Nanyang Technological University; represented the school in ECNU chess and swimming competitions, placing **5th overall and 1st among women** in chess, **4th** in breaststroke, and **5th** in freestyle.

Technical Skills

Research & Machine Learning: LLM Agents, RAG and Long-Term Memory, Multi-Agent Systems, Multimodal Learning, Trustworthy AI, Model Editing, PyTorch, Hugging Face Transformers, LLaMA-Factory, LoRA/PEFT, Distributed and Mixed-Precision Training

Programming Languages: Python, C/C++, SQL, JavaScript/TypeScript, Java, $\LaTeX$

Systems & Development: Linux, Git, Docker, Django, MySQL, Ionic Vue, Capacitor, SylixOS, EtherCAT, FlashAttention, Parallel Training, Full-Stack Development